VVB Standards & Procedures

2026

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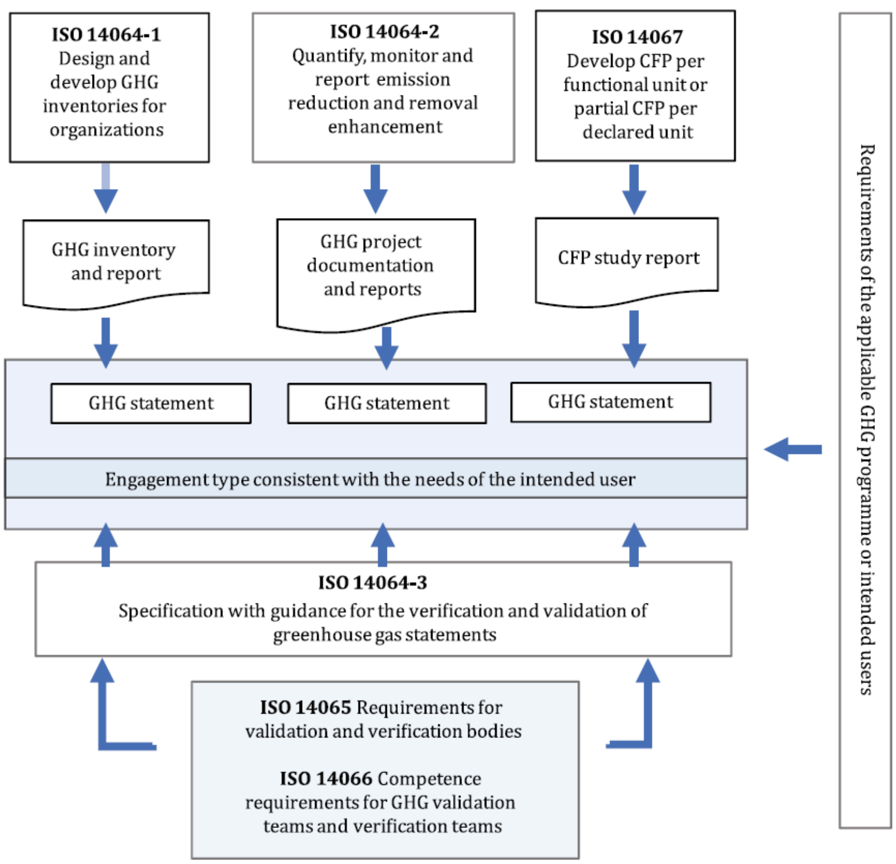
Accreditation Standards

All VVBs must meet Athian's credentialing requirements in order to perform the verification and validation functions in the Athian GHG Claim Platform. Athian approved VVBs must meet the competency requirements laid out in ISO 14065:2020 and be accredited or actively working towards accreditation by the ANSI National Accreditation Board (ANAB).



VVB Requirements - Overview

Athian will verify the VVBs meet the following requirements:

- Has obtained or is working towards ISO 17029:2019 and ISO 14065:2020 for Greenhouse Gas activities accreditation offered under the ANSI-GS Accreditation Program.
- Verifiers will be required to be certified annually in compliance with the requirements of ISO 14065 through the ANAB's accreditation program.
- Subject matter expertise in the on-farm operations related to an approved protocol (ex: Dairy Operations; Feedlot Operations)
- Experience in a particular geography where the verification will occur
- Complete onboarding in the Athian Platform
- Conduct monitoring in accordance with the requirements of the relevant protocol

Conformance to 17029:2019 and 14065:2020

Athian approved VVBs must follow the following principles and general requirements outlined in 17029:2019 and 14065:2020

Principles

- **Evidence Based Approach to Decision Making:** The process deploys a method for reaching reliable and reproducible validation/verification conclusions and is based on sufficient and appropriate objective evidence. The verification statement is based on evidence collected through an objective verification of the claim.
- **Documentation:** The validation/verification process is documented and establishes the basis for the conclusion and decision regarding conformity of the claim with the specified requirements.
- **Fair Presentation:** Validation/verification activities, findings, conclusions and statements, including significant obstacles encountered during the process, as well as unresolved, diverging views between the validation/verification body and the client are truthfully and accurately reflected.
- **Impartiality:** Decisions are based on objective evidence obtained through the validation/verification process and are not influenced by other interests or parties.
- **Competence:** Personnel have the necessary knowledge, skills, experience, training, supporting infrastructure and capacity to effectively perform verification activities.
- **Confidentiality:** Confidential information obtained or created during validation/verification activities is safeguarded and not inappropriately disclosed.
- **Openness:** A validation/verification body needs to provide public access to, or disclosure of, appropriate information about its verification process.
- **Responsibility:** The validation/verification body has the responsibility to base a validation/verification statement upon sufficient and appropriate objective evidence.
- **Responsiveness to Complaints:** Parties that have an interest in verification have the opportunity to make complaints. These complaints are appropriately managed and resolved. Responsiveness to complaints is necessary in order to demonstrate integrity and credibility to all users of verification outcomes.

- **Risk-Based Approach:** Verification bodies need to take into account the risks associated with providing competent, consistent and impartial verification.
- **Conservativeness:** When the body assesses comparable alternatives, preference is given to the alternative that is cautiously moderate.
- **Professional Scepticism:** Attitude based on recognition of the potential circumstances that can cause material misstatements in an environmental information statement.

General Requirements

- **Legal Entity:**The validation/verification body shall be a legal entity, or a defined part of a legal entity, that can be held legally responsible for all its validation/verification activities.
- **Responsibility for verification statements:** The verification body shall be responsible for, and shall retain authority for, its verification statements.
- **Management Impartiality:** The body shall ensure, through a mechanism independent of its operations, that impartiality is being achieved.
- **Liability:** The validation/verification body shall be able to demonstrate that it has evaluated the risks arising from its validation/verification activities and that it has adequate arrangements (e.g. insurance or reserves) to cover liabilities arising from its activities in each validation/verification program and the geographic areas it operates.

Structural Requirements

- The validation/verification body shall be organized and managed so as to enable it to maintain the capability to perform its validation/verification activities.

Resource Requirements

- The verification body shall have access to personnel, facilities, equipment, systems and support services that are necessary to perform its validation/verification activities.
    - Personnel assigned to verification projects must have demonstrable subject matter expertise in the on-farm operations related to an approved protocol (ex: Dairy Operations; Feedlot Operations)
- The verification body may outsource verification activities, but must maintain full responsibility for verification activities and outputs

Application & Approval Criteria

Organizations that meet the required accreditation standards or are actively working towards accreditation may apply to become an Athian approved VVB. Athian will conduct a thorough review to evaluate the applicant's qualifications, assess their eligibility, and identify the protocols that fall within the scope of their services. Applicants should be prepared to provide the following information:

- **Accreditation Certification:** A copy of a valid accreditation certification or evidence of progress toward obtaining accreditation (e.g. application and/or MOU with ANAB).
    - VVB have 24 months from the date of of the first verification project to obtain certification and must demonstrate progress during that period

- **Subject Matter Expertise:** Evidence demonstrating expertise in on-farm operations relevant to approved protocols (e.g. dairy nutrition, manure management systems, feedlot operations)
- **Regional Experience**: Evidence of experience in any particular states or regions where verification may occur
- **Team Information:** A detailed list of team members, including relevant subcontractors, with contact information, roles, and responsibilities related to VVB services.
- **Legal and Insurance Documentation:** Proof of the organization's legal status and liability insurance coverage.
- **Internal Policies and Procedures:** Examples of internal policies and procedures supporting verification activities, (e.g. conflict of interest policies, data security protocols, and data storage policies)

Athian reserves the right to request additional information or clarification during the application process to ensure the applicant's capacity to meet the required standards and expectations.

Contracting Process

Contracting for services will take place between Athian and the VVB organization. Upon application approval, the following agreements will be executed to establish the terms of the engagement between Athian and the VVB organization:

- **Mutual Non-Disclosure Agreement (NDA):** This agreement will safeguard the confidentiality of proprietary or sensitive information exchanged during the course of the engagement.
- **Master Services Agreement (MSA):** This agreement will define the overarching terms and conditions of the contractual relationship, including but not limited to roles, responsibilities, liabilities, and termination rights.

Separate statements of work (SOW) shall be established for services on a per-protocol basis. Each SOW will detail the specific services to be provided under the framework of the MSA, including:

- The scope and objectives of the services for the specific protocol
- Timelines, milestones, and deliverables associated with the work
- Fee structures, payment terms, and any additional financial considerations

Any changes or additions to the terms of any established contract must be documented through formal amendments or addenda and signed by authorized representatives of both Athian and the VVB organization.

Active Partners & Assignments

Athian will maintain a current list of Athian approved VVBs in the Athian Platform and will assign VVBs to validate protocols and verify on-farm interventions[1]. Specific project assignments will be based on the VVB's level of subject matter expertise as it relates to a particular protocol or regional expertise as it relates to the intervention location. Athian

[1] An intervention is defined as a unique implementation of a protocol on a farm.

maintains the right to revoke an assignment to a project for any reason. VVBs maintain the right of refusal for any project assigned. In order to maintain impartiality in verification, service assignments to any single intervention partner are limited to five consecutive years.

Training & Onboarding

Platform Training

The Athian platform has been designed to support the Measurement, Reporting, and Verification (MRV) process by ensuring the accurate, complete, and auditable transacting of carbon assets. Use of the platform is mandatory for all intervention and verification activities, with the exception of pilot programs.

All verifiers are required to complete a comprehensive platform training session prior to engaging in any verification activities. During this training, Athian will provide an in-depth walkthrough to demonstrate the core features and functionality of the platform, explore workflows based on real scenarios, and explain the various outcomes and reporting capabilities.

Platform training is considered complete when the verifier can:

- Navigate their assignment dashboard to view active and past projects
- Access current and historical data to perform complete desk reviews
- Add, edit, and remove issues from the issue log, specifying issue type and materiality
- View, download, and upload documents to the appropriate monitoring periods
- Complete a written assessment to provide details about the project, verification process, or outcomes
- Make a final decision that reflects the results of the verification and provide rationale
- Generate and download reports for recordkeeping

Upon successful completion of the training, user accounts will be created to grant verifiers access to the platform. Athian will provide ongoing technical support and training on future platform enhancements as necessary.

Program Guidance

Athian will establish standard methodologies for all interventions that are added to the program, while allowing for innovation and continual improvement. Consistent and repeatable processes for data collection, quantification methodologies, and verification processes will be established in the Athian platform based on the methodologies described in the Athian Governance Framework. Athian methodologies and protocols are all grounded by International Standards Organization (ISO) 14064, the Greenhouse Gas Protocol Scope 3 Standard, standards as set forth by Value Chain Initiative, the Integrity Council for the Voluntary Carbon Market and the Science Based Targets FLAG Guidance.

While no formal training on the Athian Governance Framework is provided, verifiers will be given access to all official program documentation and should be familiar with the core principles, processes, terms, and responsibilities outlined in the Athian Governance Framework.

Protocol Guidance

Verifiers may be assigned to any protocol for which they have demonstrated experience or subject matter expertise and have completed an approved statement of work. Athian will provide access to current versions of the protocol, as well as any supporting materials, such as monitoring plans, quantification tools, and documentation templates. For any newly contracted protocol, verifiers will undergo a trial period during their first intervention assignment. The trial period will take place for at least one full verification cycle, but may be extended if it is deemed necessary by either Athian or the verifier.

Verifiers should refer to the validated protocol document and Athian Monitoring Plan for specific requirements and guidance on protocol implementation, quantification, or verification, such as:

- Assessment boundaries and applicable sources, sinks, and reservoirs (SSR)s
- Eligibility criteria and qualifying practices
- Monitoring, measurement, and data collection methodologies
- Quantification methodologies for baseline and project emissions
- Uncertainty deductions and justification
- Leakage and permanence
- Causality and additionality

In instances where protocol guidance is unclear, verifiers may use their professional judgement while rendering their services, so long as it is consistent with the principles of the Athian Governance Framework and ISO standards. Deviations from protocol methodology are not permitted without express approval of a Request for Variance. Requests must be filled out by the producer or implementation partner and approved by both the verifier and Athian.

Verifiers are encouraged to contact Athian for additional clarification, guidance, or feedback on any protocol.

Dose Verification Guidance

The following guidance has been developed to support verification of feed additive/ingredient protocols in the event that variation occurs between the formulated dose and the actual dose.

1. **For protocols where dosage does affect the final reductions:** if the calculated reduction remains within the 95% level of accuracy detailed in the protocol, the dosage variation is acceptable, as long as the producer can demonstrate that the formulated dose met the eligibility requirements of the protocol.
2. **For protocols where dosage does *not* affect the final reductions:** the published scientific evidence that was reviewed during the protocol approval process for both the lowest possible effective dose and highest studied dose will define the upper and lower limits of the acceptable actual fed dosage range, as long as the producer can demonstrate that the formulated dose met the eligibility requirements of the protocol. These upper and lower limits will be identified in the monitoring plan for each protocol.
3. **If the second rule conflicts with the first** (e.g. a situation where the actual dosage falls below the scientifically supported efficacy floor – or above the maximum studied dose – but still results in a calculated reduction within the 95% level of accuracy), the

scientifically supported boundaries will define the acceptable actual dosage limitations.

Protocol Pilots

As part of the protocol approval process, each protocol must successfully complete a pilot before it is cataloged and made available at scale on the Athian platform. The pilot phase is conducted with a select group of producers and verifiers to evaluate the performance of the protocol and develop necessary supporting materials under real-world conditions. Athian will select VVBs to participate in pilots based on their experience and subject matter expertise as it relates to the protocol being tested.

Throughout the pilot phase, Athian will meet with the participants on a regular cadence to:

- Better understand any protocol-specific nuances in the verification process
- Develop and refine materials to provide guidance for implementation and verification
- Gather feedback to enhance the platform's functionality and support for the protocol
- Identify common challenges or barriers that may arise and address them prior to scaling

This rigorous pilot process is essential to ensuring the success of each protocol, enabling the development of consistent and repeatable procedures at scale. Verifiers who would like to participate in a protocol pilot are encouraged to contact Athian for consideration.

Ongoing Verifier Training

As part of a regular review, verifiers working with Athian will receive an annual mock audit for each of the protocols they are approved to verify within the Athian platform. This mock audit will take place prior to the Protocol Annual Review process to incorporate learnings from these audits into improvements and re-training needs for each protocol on a yearly basis.

Verifiers will be asked to provide supporting documentation addressing the following areas for each of the interventions they verify. In addition, verifiers must include evidence detailing the methodologies and processes that informed their approach and any tools or resources used, where applicable. Responses to the following areas should also include a qualitative description of the specific team processes employed by the organization:

1. **Protocol Adherence**
   Description of the process used to determine that each project adhered to the applicable protocol.

2. **Site Visit Verification**
   Evidence of site visits conducted, including visit dates, personnel involved, and key findings. In addition, provide the methodology used to perform the site visit(s).

3. **Risk Analysis**
   A summary of the risk assessment process applied to each project, along with any identified risks.

4. **Non-Conformities**
   Identification of any non-conforming issues and the actions taken to address or remediate them.

5. **Review Processes**
   Overview of your primary review and peer review procedures, including the roles and qualifications of reviewers. Provide information on how those roles are kept separate/unbiased.

6. **Eligibility and Compliance**
   Explanation of how producer eligibility and regulatory compliance requirements were assessed and confirmed.

7. **Verification Planning and Methodology**
   Description of the planning process and the specific methodology applied in verifying the projects.

8. **Data Sufficiency**
   Justification for how the documentation and data collected were deemed sufficient to support verification outcomes.

9. **Standards Alignment**
   Explanation of how the verification results map back to relevant ISO standards.

10. **Level of Assurance**
    Rationale for determining whether the results represent a Limited or Reasonable level of assurance.

11. **Independent Tools and Resources**
    Disclosure of any tools or resources used outside of the Athian platform to support quantification or validation of project results.

Verification Engagements

Conflict of Interest

When conducting verification under this protocol verifiers must be seen as credible, independent, and transparent. To meet this requirement, a conflict of interest (COI) determination must be made prior to starting any verification activities. A COI occurs in any situation that compromises the verifier's ability to perform an independent verification.

Every verifier must provide information about its organizational relationships, internal structures, and management systems for identifying potential COIs. Verifiers must evaluate any potential conflicting services it has provided to the project developer, including any advice or consulting provided outside of the verification process.

Verification Objectives & Standards

The objective of verification is to verify that an assertion for GHG emission reduction is materially accurate and in conformance with the specific requirements of the protocol. The verification body must be able to verify that all eligibility requirements were met, quantifications are accurate and free of material discrepancies, and the methodologies described in the protocol were applied without deviation. The quantitative threshold for materiality is set at a >95% level of accuracy. This means that the project's calculated emission reductions must be less than 5% different than those calculated by the verifier.

The verification body must provide a factual statement expressing the outcome of the verification, either limited or reasonable. Issues identified during verification must be classified by verification bodies as either material (significant) or immaterial (insignificant). To be verified successfully, all reported emissions reductions must be free of material misstatements.

For the context of verification, Athian defines reasonable assurance and limited assurance as follows:

Reasonable:

- Final delivered documents accurately and completely reflect reported values.
- Final claim data elements have been determined to be free of material reporting or mathematical errors as defined above.
- High confidence that all verified elements (data and documentation) are representative of the actual implementation of the intervention as occurred on farm during the monitoring period.
- A physical site visit has been performed as part of the initial verification of the first monitoring period or as part of a risk-based assessment of the implementation of the intervention on farm in subsequent monitoring periods.

Limited:

- Documentation is free of material reporting or mathematical errors.
- Alternative forms of documentation may have been accepted due to the inability of the farm to provide the highest standard of documentation for certain data elements.
- Any issued and documented variances, to the monitoring period or protocol itself, are present for the monitoring period.
- High confidence that as reported, the claim is free of material error.
- A site visit was performed as part of the initial verification of the first monitoring period, but took place virtually.

Verification Process

To verify the project, the verifier must develop a risk-based verification plan that takes into account the size and complexity of the operation. The verifier must follow the following process:

1. A conflict-of-interest evaluation
2. A verification plan that includes, at a minimum:
    a. The lead verifier and independent reviewer
    b. A list of the location and dates of any site evaluations that will be conducted
    c. The types of data and documents that will be reviewed by the verifier
    d. A list of the people who are expected to be interviewed as a part of the verification

3. A kick-off meeting with all parties to lay out the timeline and process of the verification
4. At minimum, one annual site visit to confirm practice implementation and operational activities relevant to the applicable protocol(s). This site visit may occur virtually or with a physical in-person visit, per the definitions of Reasonable and Limited Assurance in Verification Objectives and Standards.
    a. Required procedures for all site visits:
        i. Determine with the producer an acceptable date and time for the visit. Reference the applicable Monitoring Plan for visit timing requirements.
        ii. Share with the producer an agenda for the meeting.
        iii. Share with the producer required and optional participants for both parties.
    b. Required procedures for physical site visits:
        i. Confirm any biosecurity or safety procedures in place prior to arrival.
    c. Required procedures for virtual site visits:
        i. Prior to the meeting, verifiers should request an annotated site map to pre-determine areas of required visual evidence.
        ii. The producer must be able to visually present the activities of the intervention. This may be accomplished three ways:
            1. Videographic evidence, either live or recorded with a date and timestamp to ensure relevance.
            2. Photographic evidence with a date and timestamp to ensure relevance. Verifiers should encourage annotations and/or captions that describe what is visible in each image.
            3. Satellite evidence with a date and timestamp.
        iii. The producer should be prepared to either share their computer screen or allow for a team viewer function to demonstrate the original source of reported data/documentation. This should originate from an on-farm software system wherever possible.
    d. Verifiers may perform virtual site visits to confirm the activities associated with the protocol to satisfy the site visit requirement. A physical site visit should be conducted in the following circumstances, in consultation with Athian:
        i. If the assigned VVB determines that the intervention circumstances necessitate a site or facility visit based upon the results of a risk assessment, evidence gathering plan, prior verification results at the same site/facility, and/or the circumstances defined in ISO 14064 Part 3 Site Visit Requirements & Risk Variables (listed below).
            1. A verification where there has been a change of ownership of a site or facility and where the emissions, removals, and storage on the site or facility is material to the GHG Statement.

a. The addition of a site or facility that is material to the GHG Statement.

2. When misstatements are identified during the verification and indicate a need for a site visit.

3. There are unexplained material changes in emissions, removals, and/or storage since the previous verified GHG statement.

    a. Material change in the scope or boundary of reporting.

        i. Including significant growth of participating animals, change in equipment that materially affects the activities of the protocol (i.e. new manure management equipment, the installation of a micromachine, etc.), or other significant operational changes.

    b. Significant change in the data management involving a specific site.

5. A desk review of the data from the project

6. A verification report that includes:

    a. A verification statement documenting the outcome of the assessment (reduction results) and if there was any material discrepancy noted

    b. Key details about the project including: producer and farm operation identification, verifying body and lead verifier contact information, protocol information, and intervention information

    c. A description of the protocol, the objectives and criteria used to arrive at the final result, the scope of the project, the level of assurance associated with the project, and any details about the implementation of the practices observed

    d. Detail about the verification process used to complete the assessment including approach and methods and also noting any conflict of interest

    e. Verification findings including confirmation of producer eligibility, adherence to the criteria established in the protocol, the verified emissions quantification values, and the final written opinion of the verifier(s)

    f. An issue log capturing any issues identified during the verification and their classification as either material (significant) or immaterial (insignificant)

    g. A representation of all data & documents used in the process of verification

Claim Generation & Registration

Athian will engage third-party Validation and Verification Bodies (VVBs) to determine if protocols are in compliance with International Organization of Standardization's ISO 14064-2 (specifications and guidance at the project level for quantification, monitoring, and reporting of greenhouse gas emission reductions or removal enhancements).

Each implementation of a protocol (i.e. intervention) will require verification by an Athian approved, independent, third-party verifier. The verifier determines if each producer has implemented the intervention in a compliant manner and if the value of the reduction is quantified accurately in accordance with the protocol.

Once validation of a protocol and verification of a quantified reduction have been completed, a claim can be certified and recorded in the Athian System of Record. Athian uses the term AVSA (Athian Verified Sustainability Asset) to refer to certified reduction claims generated through the Athian Platform. Each AVSA is given a unique identification number corresponding to the month of its generation. This unique identifier enables Athian to track each claim throughout sale and retirement to ensure no double-counting.

Protocol Approval Process

All interventions that take place on the Athian Platform will be verified against approved protocols for quantifying GHG emission reductions and removals. These protocols provide a consistent, standardized approach for producers to implement interventions and for verifiers to assess the results of those efforts.

Protocols must complete the following process in order to be added to the Athian Platform:

1. **Outline Submission:** Intervention Proponents will submit an overview of a possible protocol to Athian.
2. **Outline Review:** Athian will review to determine if the suggested protocol meets the requirements for the Athian program. If the suggested protocol meets Athian eligibility criteria, the Intervention Proponent will draft a "Proposal for Protocol" to be submitted to the SAB. The "Proposal for Protocol" requirements are outlined below.
3. **Scientific Review:** The SAB will review a "Proposal for Protocol" to determine if the science supporting the proposed protocol is credible. The SAB will evaluate for:
    a. **Executive Summary:** Overview of why this intervention will have a positive impact on GHG emissions within the supply chain.
    b. **Eligibility:** Description of what livestock operations are eligible to participate in this protocol, what regions are applicable, and any other qualifying criteria.
    c. **Causality:** Demonstration that an investment (or other equivalent action) of a company or group of companies acting collectively is what caused the Intervention to happen. Define how causality will be determined for this protocol.
    d. **Implementation Barriers:** Define how the intervention will remove barriers to reducing emissions within the value chain.

   e. **Permanence & Leakage:** Address permanence & leakage, and supporting science, if applicable. Describe if there is a risk of reversals and if so, how would it be mitigated.
   f. **Emissions Boundaries:** Clearly defined boundaries used in the quantification.
   g. Brief Methodology:
      i. Describe if the calculations are absolute or intensity based.
      ii. Ensure the research is current and relevant to supply chain conditions.
      iii. Provide the actual calculations to be used to quantify the impact of this intervention on GHG emissions and ensure the methodology supports the expected results and is based on the references cited.
      iv. Describe any uncertainty in the data and how to mitigate risks.
   h. **References:** Any studies, articles, publications, or other supporting literature.
4. **Draft Protocol:** Intervention proponents will draft a protocol in accordance with Athian's governance program standards.
5. **Protocol Review:** Athian reviews the protocol to ensure it is in compliance with Athian's protocol requirements.
6. **Validation Review:** Protocol is reviewed for compliance with ISO 14064-2 by an accredited third-party validator.
7. **Piloting:** The protocol is piloted in the Athian platform by an Athian selected group of producers and VVBs. Piloting allows Athian to pressure test new protocols before scaling them out to a large number of producers.
8. **Verification Review:** The third-party verifier reviews the results to verify that protocol requirements were met and the reduction was quantified accurately.
9. **Cataloging:** The protocol is added to the catalog of fully piloted protocols and made available for public review.
10. **Protocol Rollout:** The protocol is made available to all applicable producers in the Athian platform.
11. **Annual Review:** The protocol is reviewed by the SAB on an annual basis to determine if the science supporting the methodologies is still sound.
12. **Continue, Revise or Retire Protocol:** The protocol is either continued, revised and re-validated, or retired. Protocol version numbers are updated in the Athian platform, if applicable.